Assignment 2

Cloud Computing Architecture
COS20019

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I. Introduction

This report outlines the creation of Highly Available Environment for Photo Album website, the project is an extension on previous assignment 1b. The Photo Album website allow user to upload photos with its meta-data such as photo tile, description, creation date, and keywords, then the website can display those photos and meta-data to user. The website will be hosted on EC2 web servers which are in private environment and could communicate with internet utilizing Elastic Load Balancer (ELB) and Network Address Translation (NAT) gateway. AWS Auto Scaling Group (ASG) is used to create the highly available environment that is scalable based on demands. Lambda function will resize the photo to create a smaller thumbnail, which will be triggered when new photo had been uploaded to S3 and resized thumbnail will be stored in S3 as well.

To archive this final product, S3 bucket policy need to be update, Dev Server instance will be used to create AMI for Web Server launch template that will be used in auto scaling group. An exists IAM roles on learner lab will be applied to Lambda Function, Dev Server instance and Auto Scaling Group.

II. Infrastructure Implementation

A. *Virtual private cloud (VPC)*

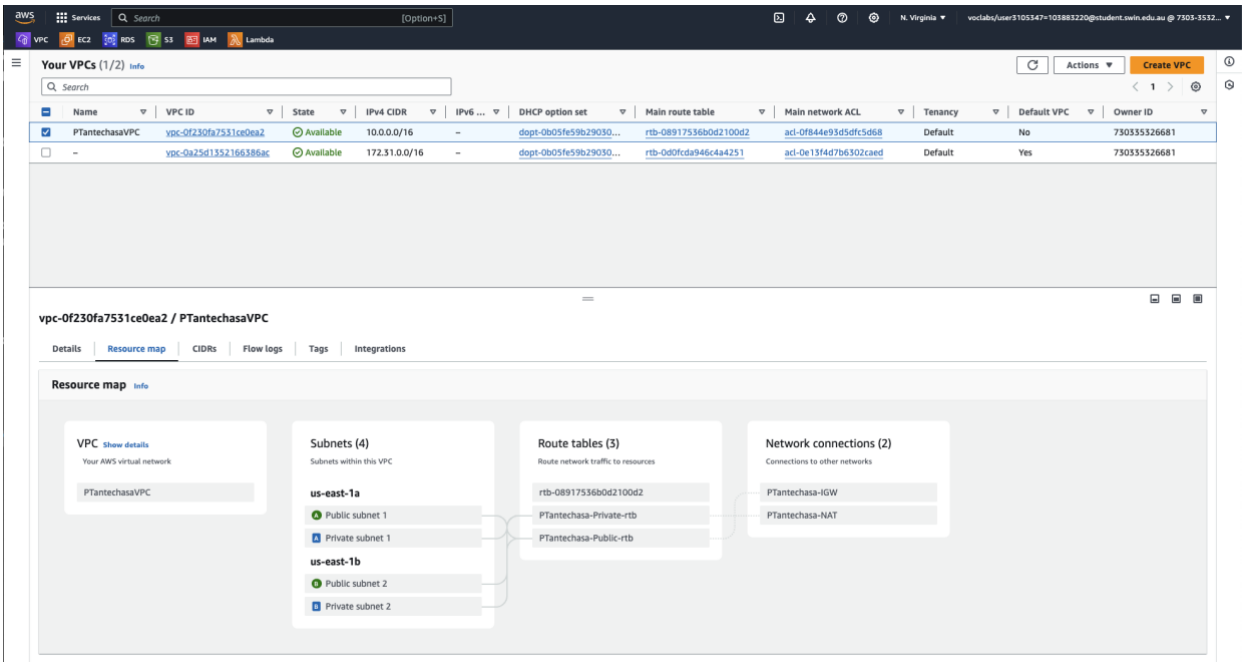

Figure 1: VPC Overviews

Name	***IPv4 CIDR***	***Availability Zone***
Public Subnet 1	10.0.1.0/24	US-EAST-1A
Private Subnet 1	10.0.3.0/24	US-EAST-1A
Public Subnet 2	10.0.2.0/24	US-EAST-1B
Private Subnet 2	10.0.4.0/24	US-EAST-1B

B. Routing Tables

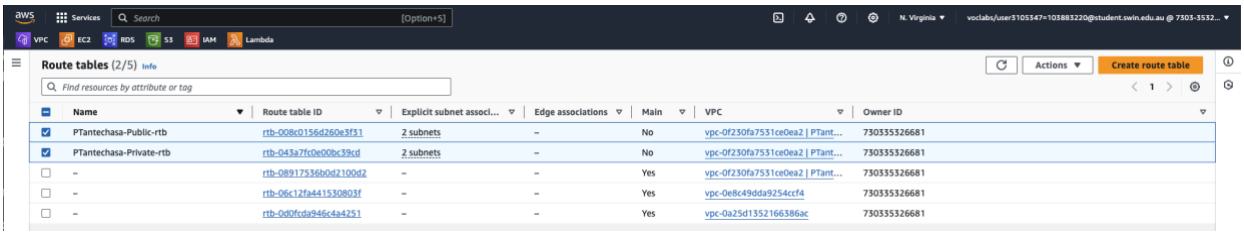

Figure 2: Routing Table

Name	***Explicit Subnet Associations***	***Routes***
PTantechasa-Public-rtb	- Public Subnet 1 - Public Subnet 2	Internet Gateway
PTantechasa-Private-rtb	- Private Subnet 1 - Private Subnet 2	NAT Gateway

C. Security Groups

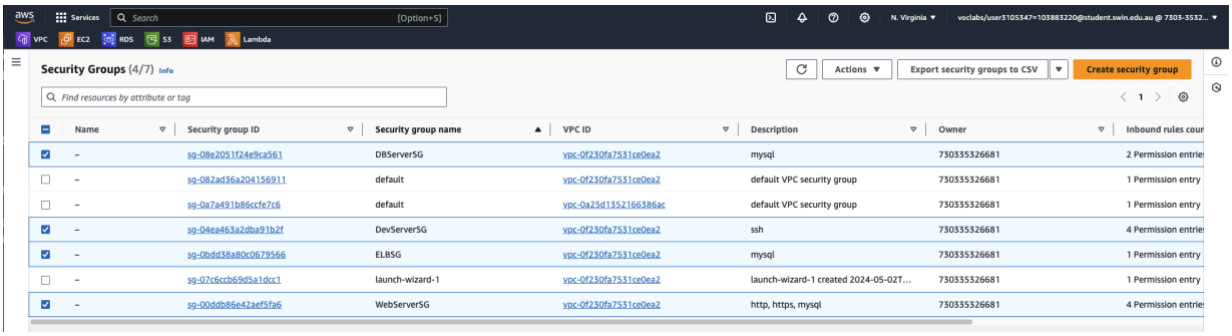

Figure 3: Security Groups Overview

a. ELBSG

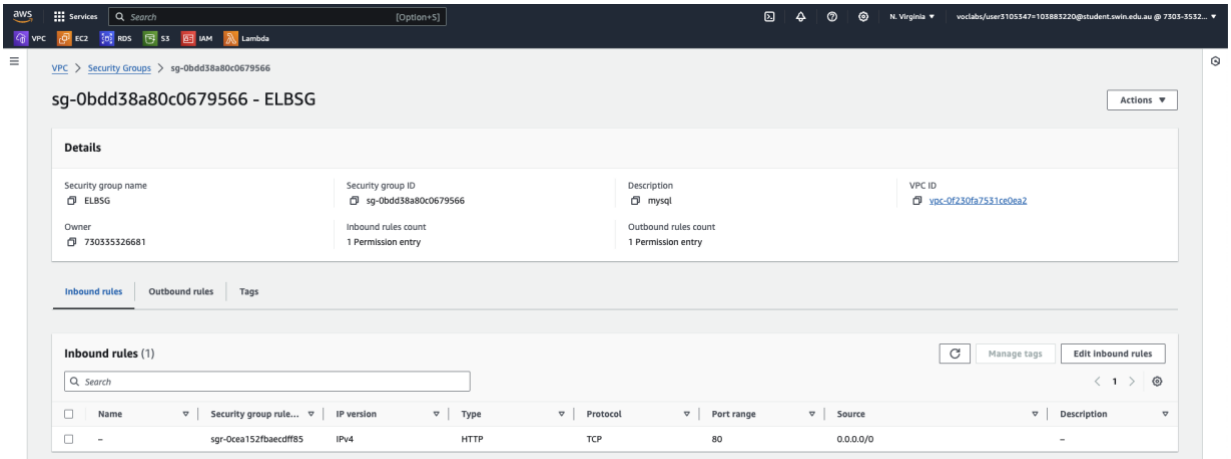

Figure 4: Security Group for ELB, Allow inbound HTTP from anywhere and outbound to Web Server

b. *WebServerSG*

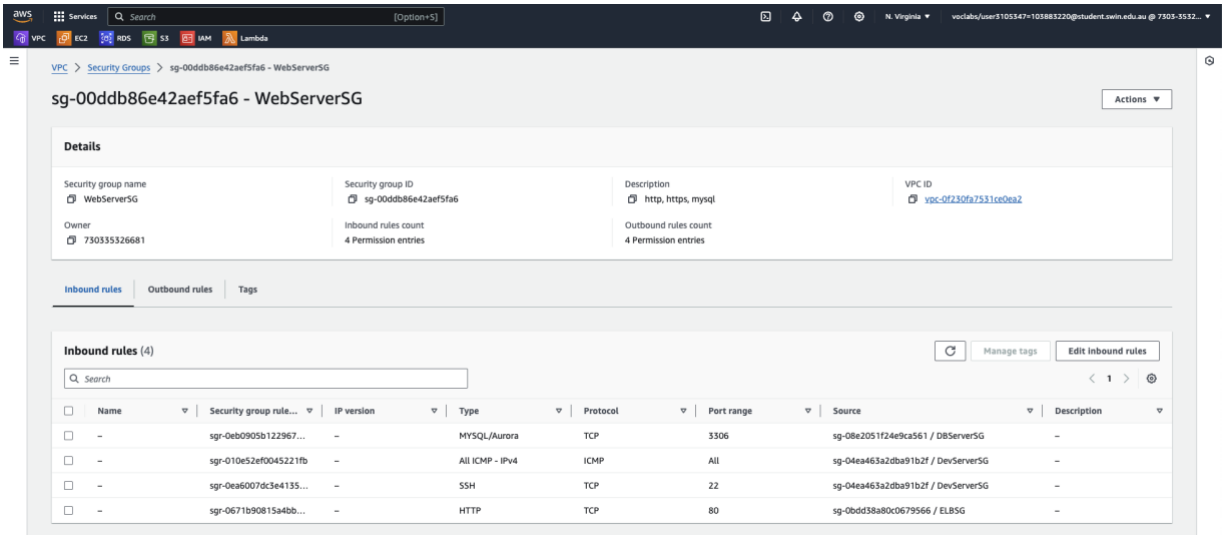

Figure 5: Security Group for Web Server

Allow inbound HTTP from ELB, SSH and ICMP from Dev Server, and MySQL from Database. Allow outbound HTTP and HTTPS to anywhere, SSH and ICMP to Dev Server, and MySQL to Database. The reason to allow to ICMP from Dev Server is to demostrate that NACL security that been configured to deny ICMP communication from Dev Server is working. (more info on *Figure 7*)

c. *DevServerSG*

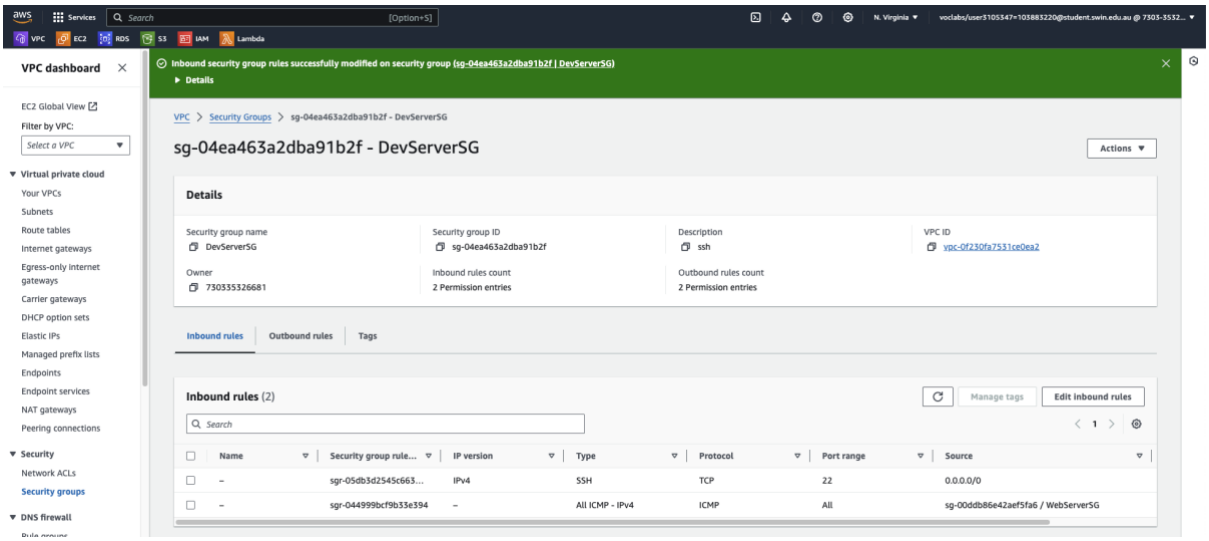

Figure 5: Security Group for Dev Server

Allow inbound SSH from anywhere and ICMP from Web Server. Allow outbound SSH to anywhere and ICMP to Web Server. At the implementatio stage, Dev Server was allowing communicaiton through HTTP and MySQL to setup and test funcitonality of the website, and had been removed to only allow website to be access from ELB and Web Server.

d. DBServerSG

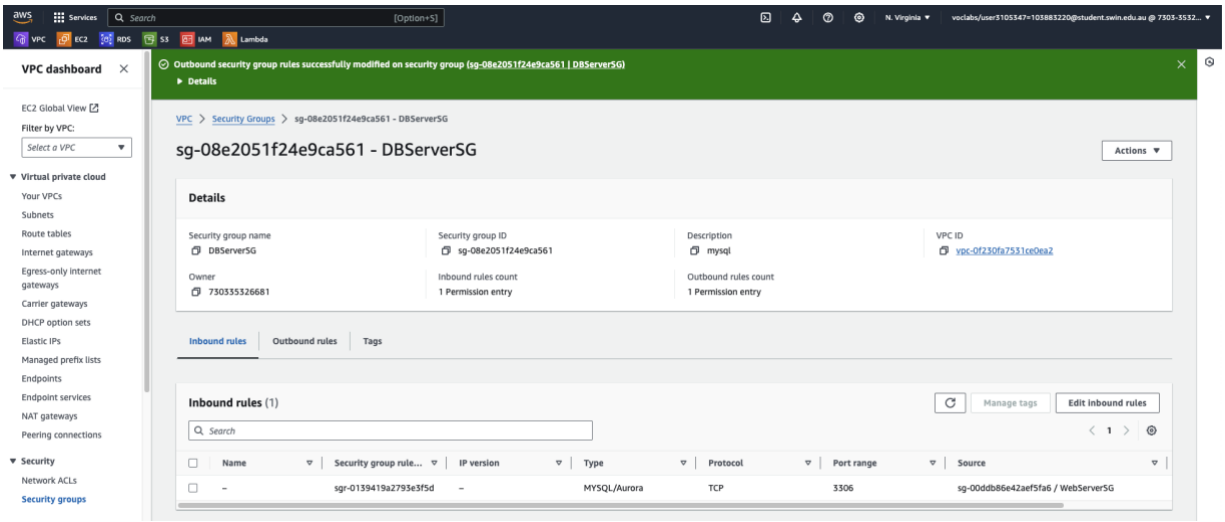

Figure 6: Security Group for Database, this only allow MySQL communication between Web Server.

D. Network ACL

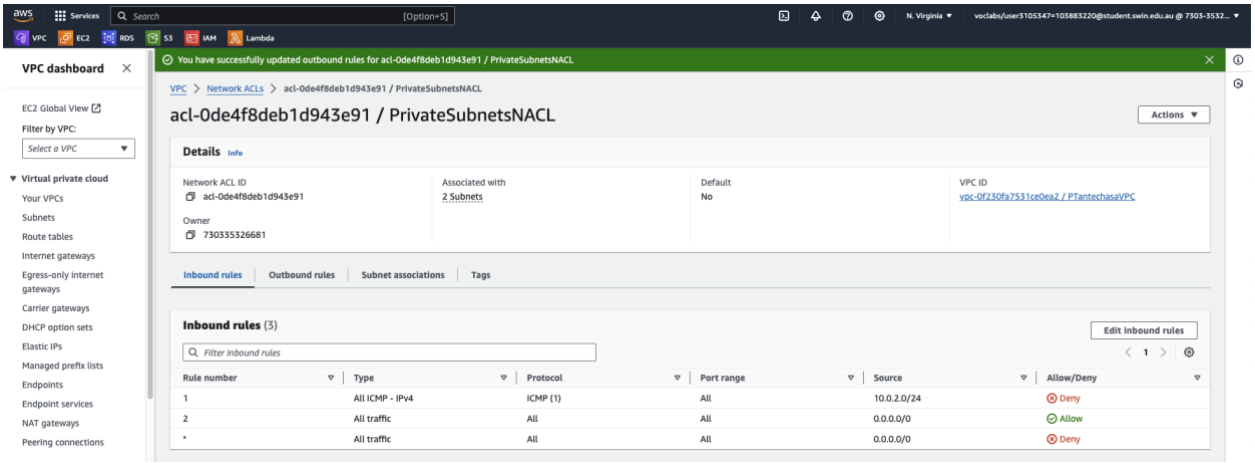

Figure 7: NACL Configuration

PrivateSubnetsNACL is applied to both private subnets, it job is to deny all ICMP communication from Dev Server even though the Web Server security group is allowing ICMP communication from Dev Server. (Test result is shown in *Figure 28*)

E. NAT Gateway

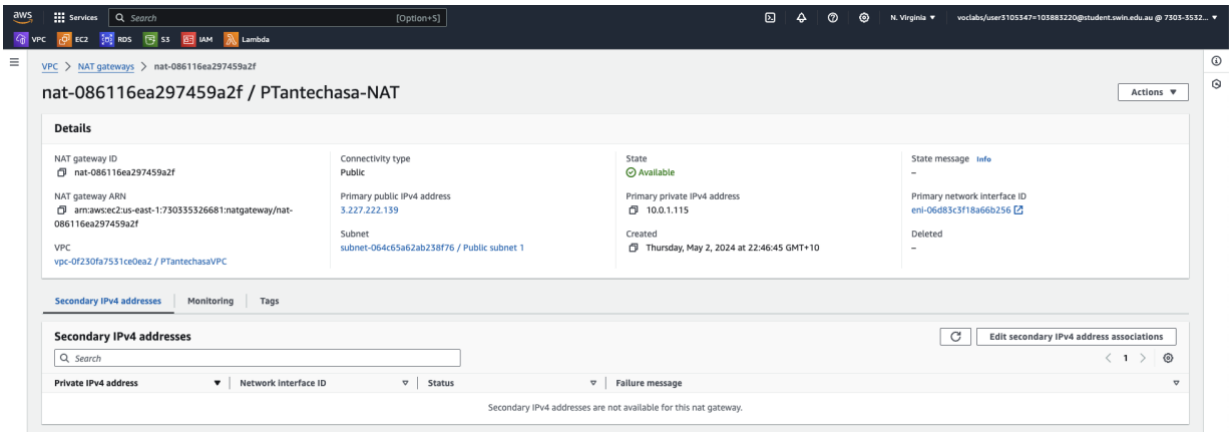

Figure 8: NAT gateway, located in Public subnet 1, allowing communication between Web server and Internet.

F. EC2 Instance

a. Dev Server

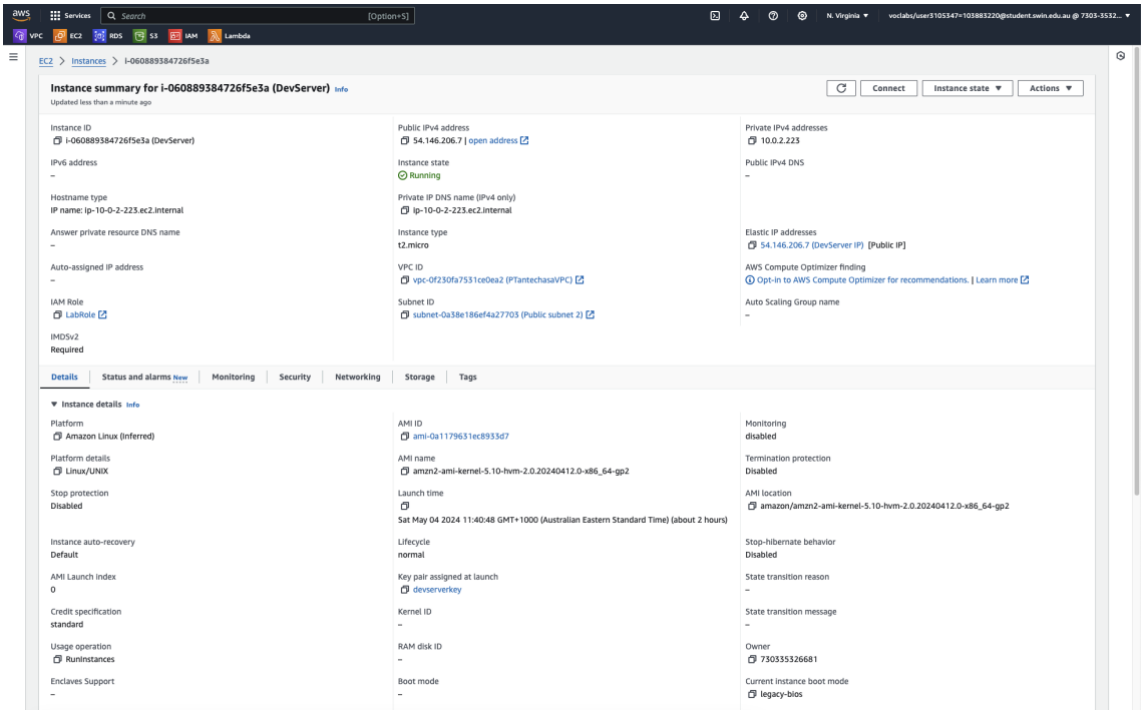

Figure 7: Dev Server Configuration

Dev Server is used for AMI creation for Web Server, AWS SDK and PhpMyadmin was downloaded and unziped, also uploaded provided photo album website with modified information to match this environment including bucket name, database endpoint, database table name and lambda function. LabinstanceRole is also applied in this step as well.

b. Web Server

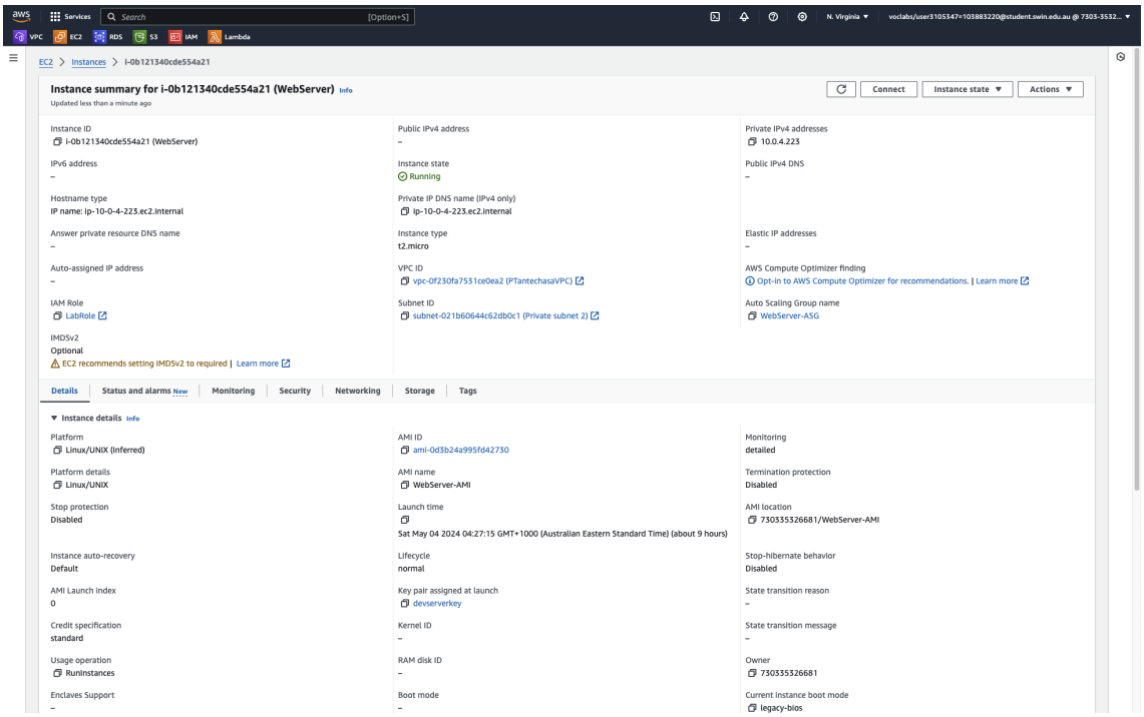

Figure 8: Web Server Configuration

Web Server is a instance created with auto scaling group using AMI created from the Dev Server. This will be receive communication from internet through ELB and send out communication through NAT gateway. This will be only place to allow user to access the photo album website.

G. Highly Available Environment

a. AMI

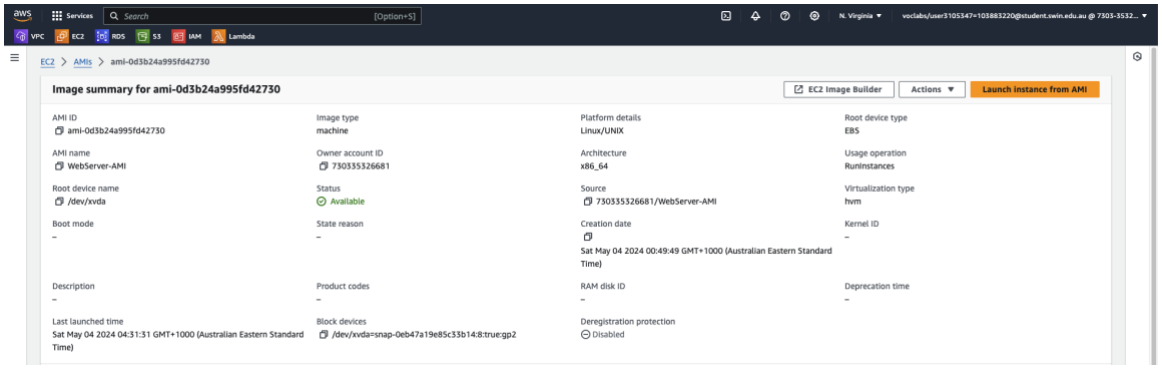

Figure 9: WebServer-AMI details

b. Target Group

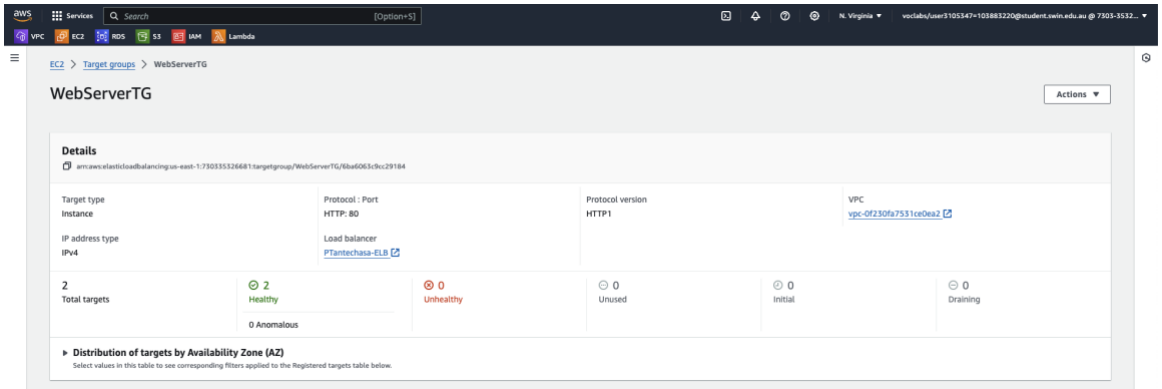

Figure 10: WebServerTG details

c. Elastic Load Balancer

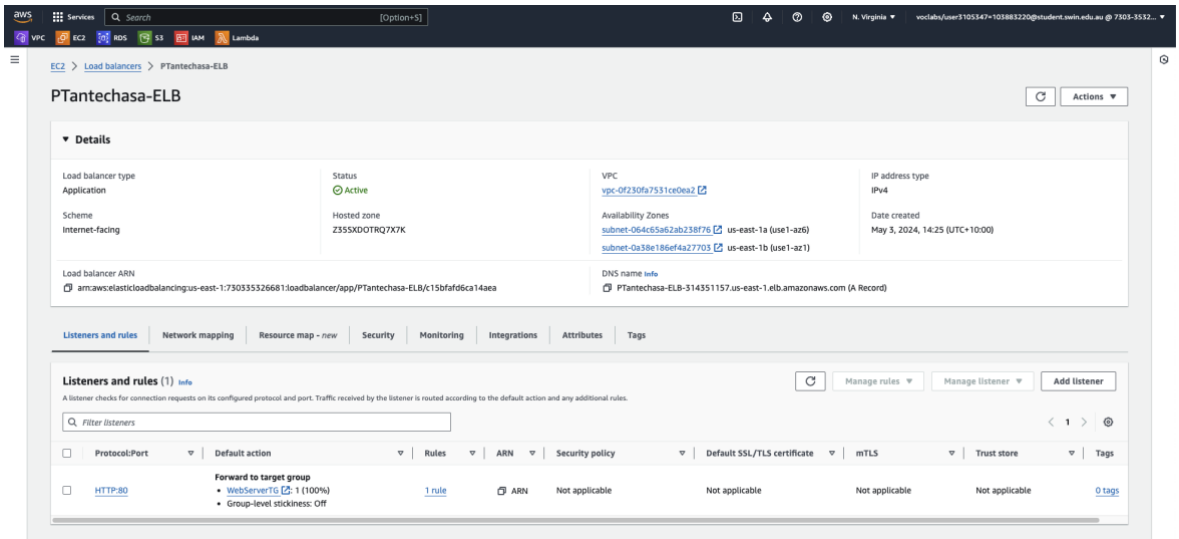

Figure 11: PTantechasa-ELB details

d. *Launch Template*

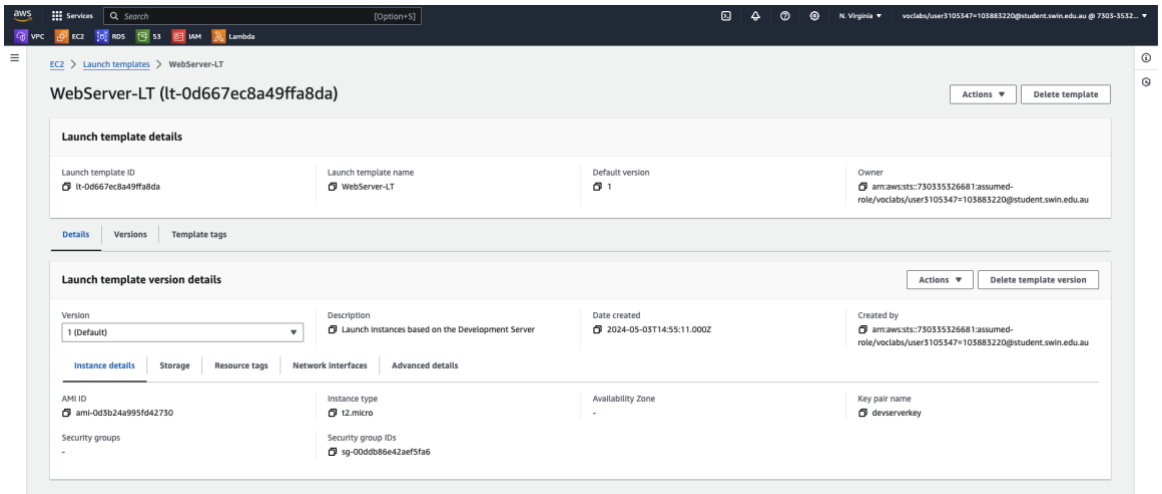

Figure 12: Launch Template details

e. *Auto Scaling Group*

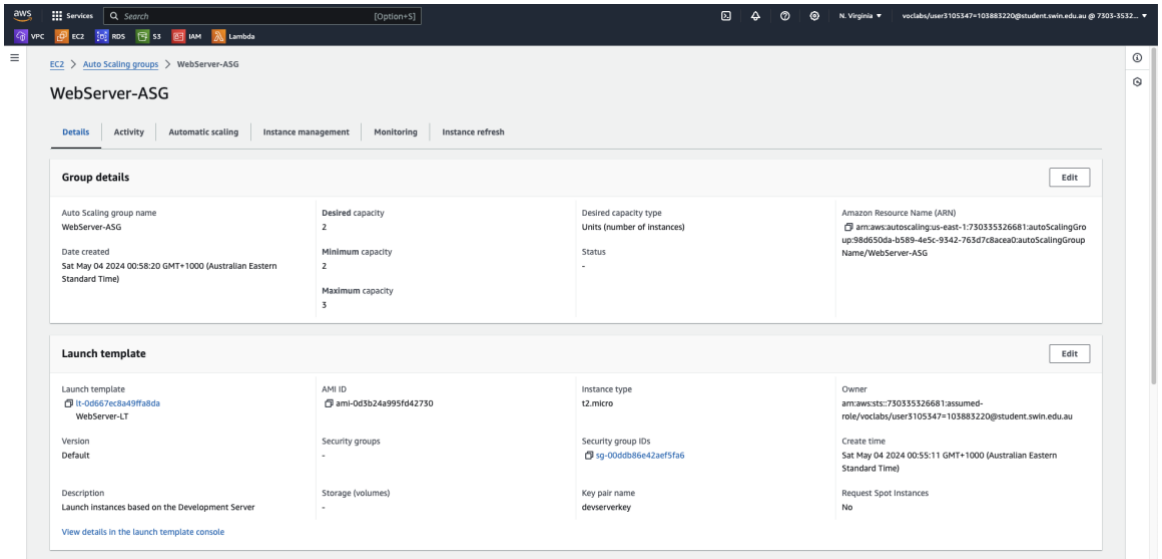

Figure 13: WebServer-ASG details, auto scaling group created with the capacity of 2 desired, 2 minimum and 3 maximum

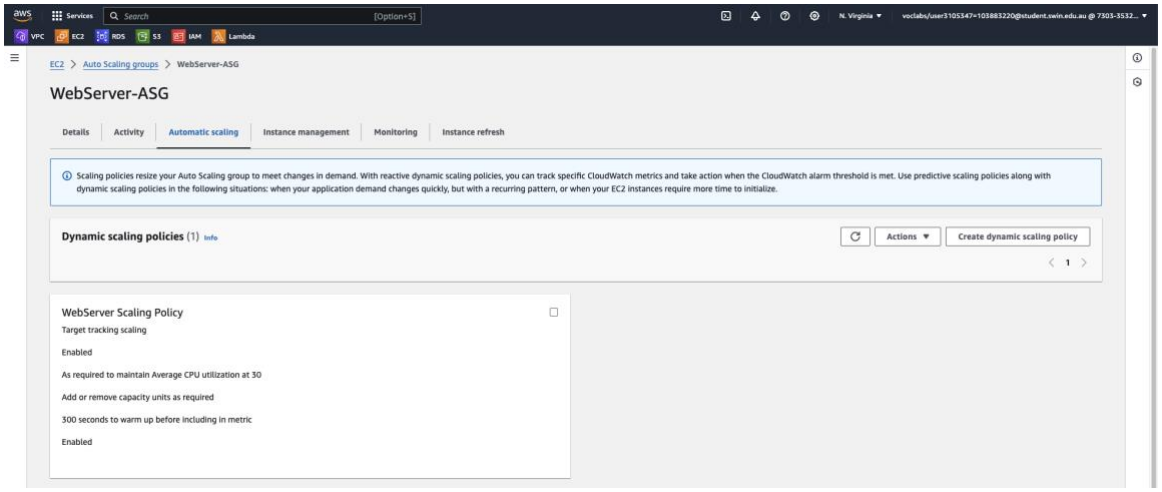

Figure 14: WebServer-ASG scaling policy, target CPU utilization at 30%

H. RDS

a. Database

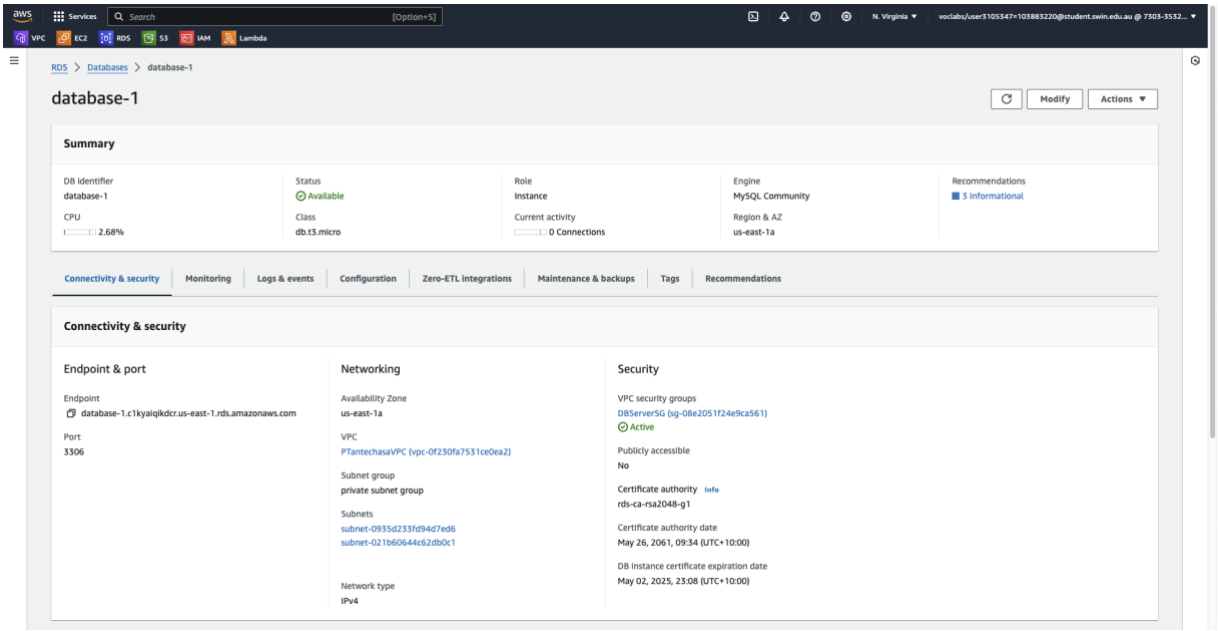

Figure 15: Database details, stored meta-data and linked to phpmyadmin

b. PhpMyAdmin

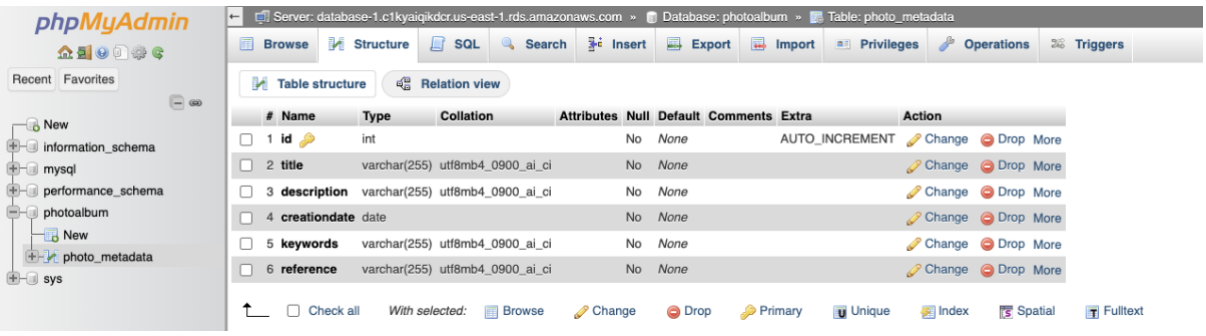

Figure 15: PhpMyadmin configuration, used for database and table structure creation

I. S3 Bucket

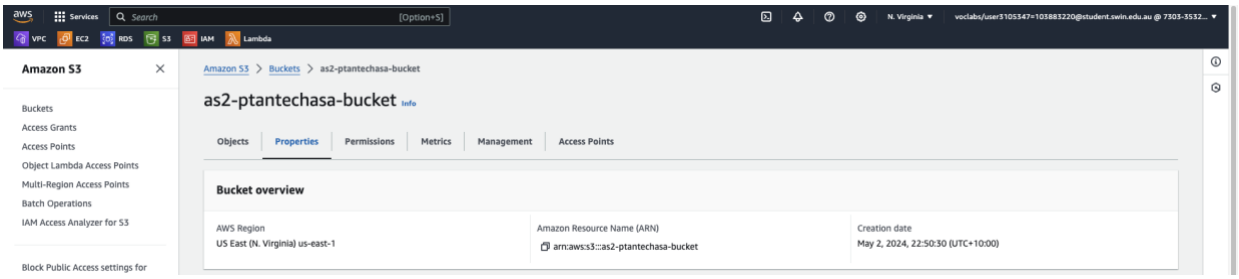

Figure 16: S3 Bucket details, used to store original photo and resized photo files that's been uploaded from the website

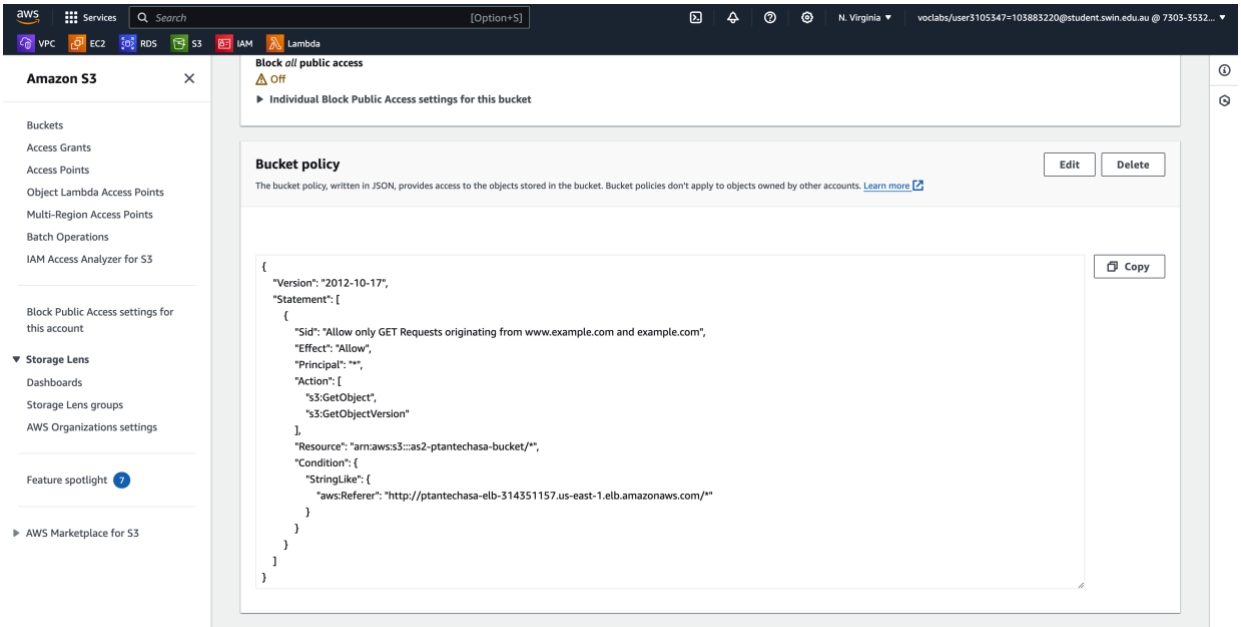

Figure 17: Updated S3 Bucket policy. Only allow GetObject from request that originated from the ELB.

J. Lambda Function

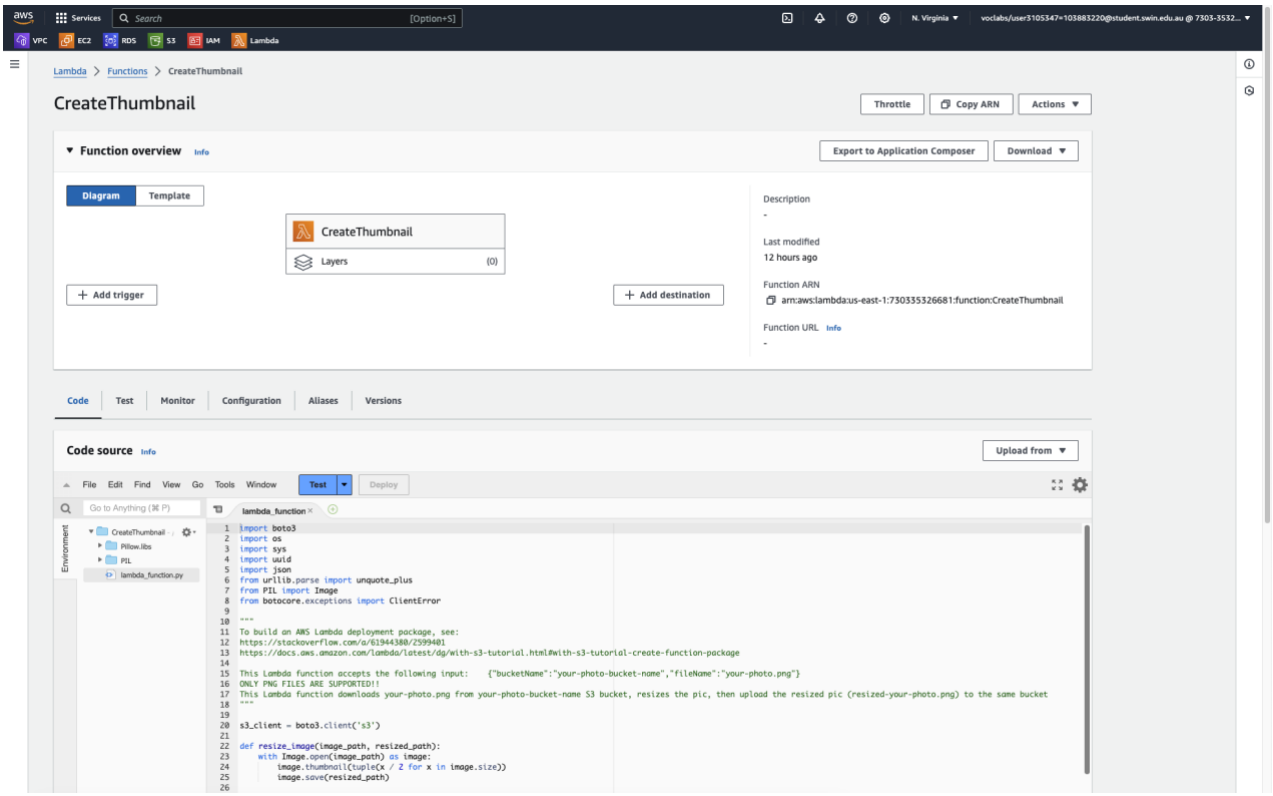

Figure 18: Lambda Function details
Triggered when photo had been uploaded to S3 bucket, download that photo, resized and reupload to the bucket.

K. IAM Role

Evidence of IAM roles that been applied multiple components for this project.

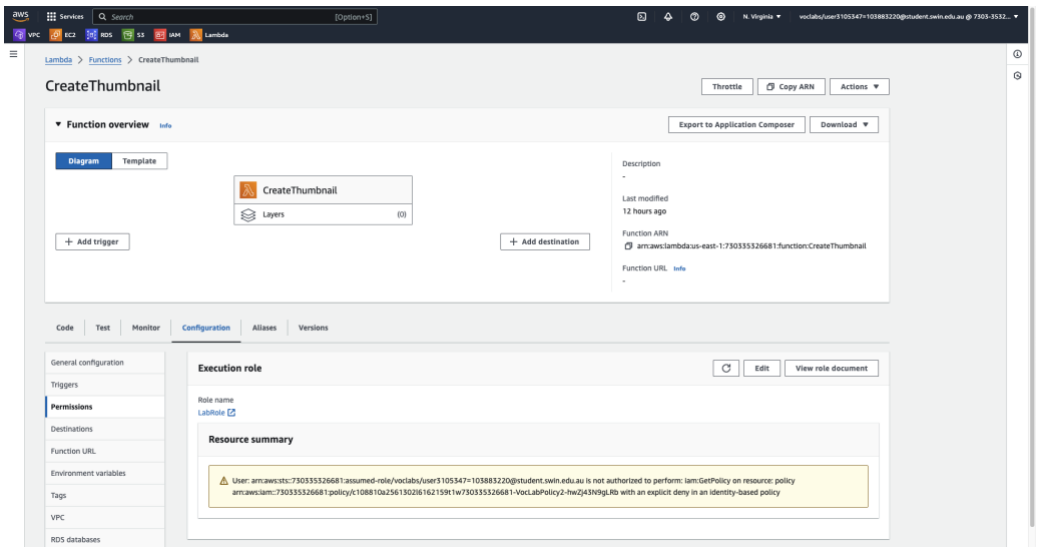

Figure 19: IAM roles applied for Lambda Function

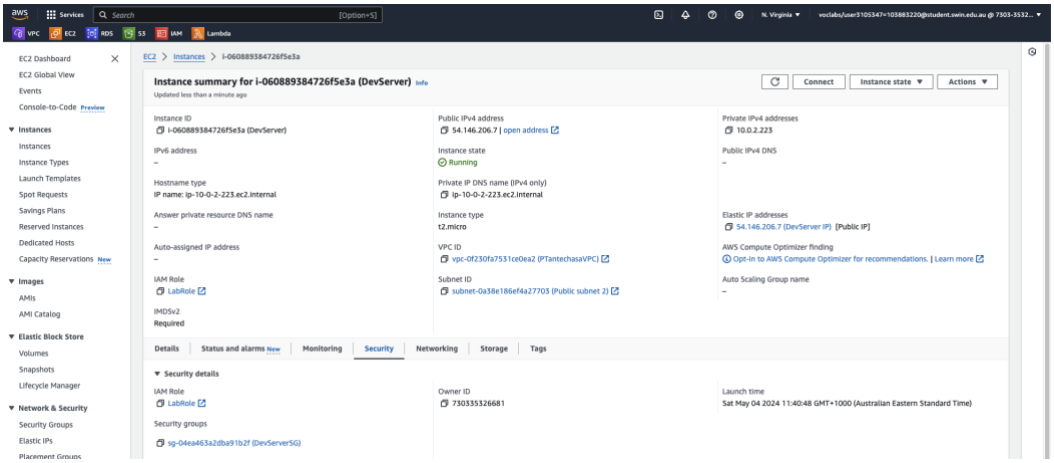

Figure 20: IAM roles applied for Dev Sever

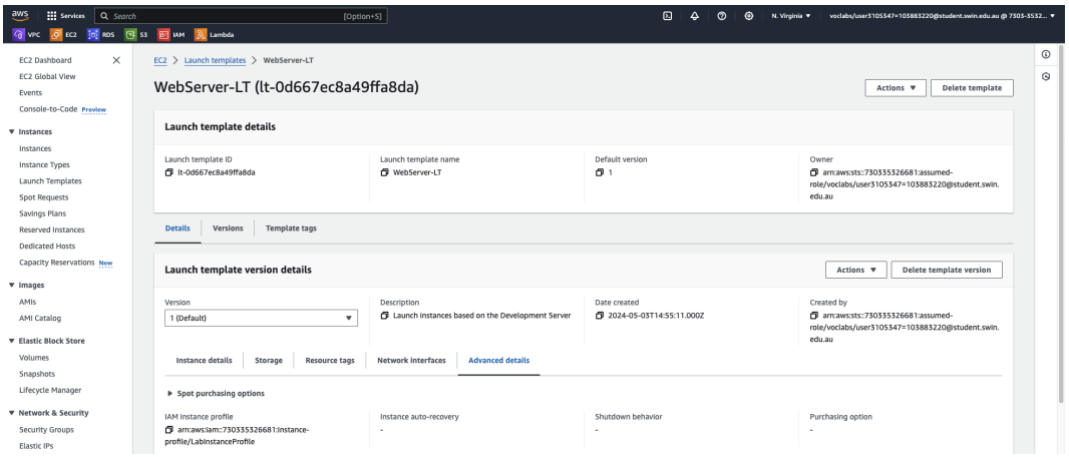

Figure 21: IAM roles applied for Launch Template

III. FUNCTIONALITY TESTS

A. PhotoAlbum Functions

URL: http://ptantechasa-elb-314351157.us-east-1.elb.amazonaws.com/photoalbum/album.php

Photoalbum website is working correctly, the user can upload a photo and its meta-data to S3 bucket and RDS database through 'photouploader.php' and correctly display on 'album.php'.

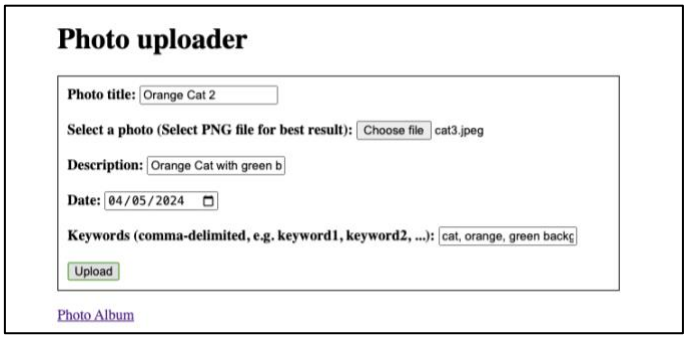

Figure 22: Photo uploader web page, allow user to upload photo to the S3 and database

Photo	Name	Description	Creation date	Keywords
	Cat	cat	2024-05-03	cat, small
	Orange Cat	Orange Cat	2024-05-04	cat, orange
	Orange Cat 2	Orange Cat with green background	2024-05-06	cat, orange, green background

Figure 23: Album web page, display photo and its meta-data to user

B. S3 Bucket

a. Photo Uploaded

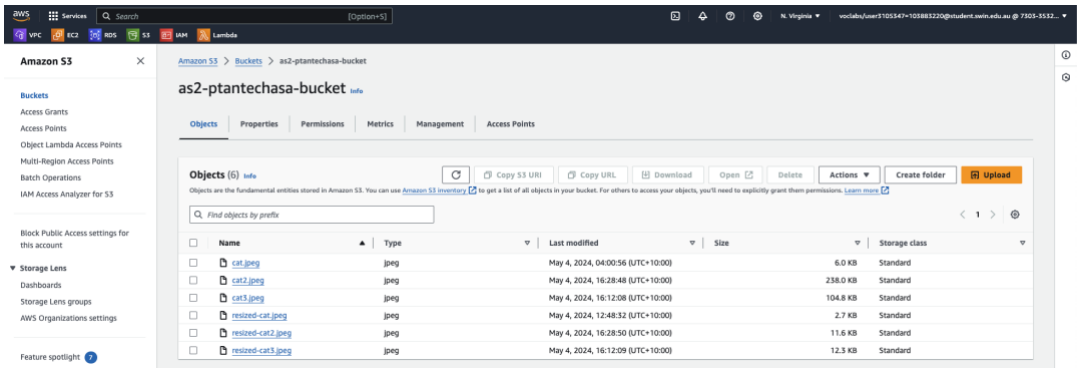

Figure 24: S3 objects, shown that original and resized photo is successful uploaded

b. Denied Direct Access to S3 Photos

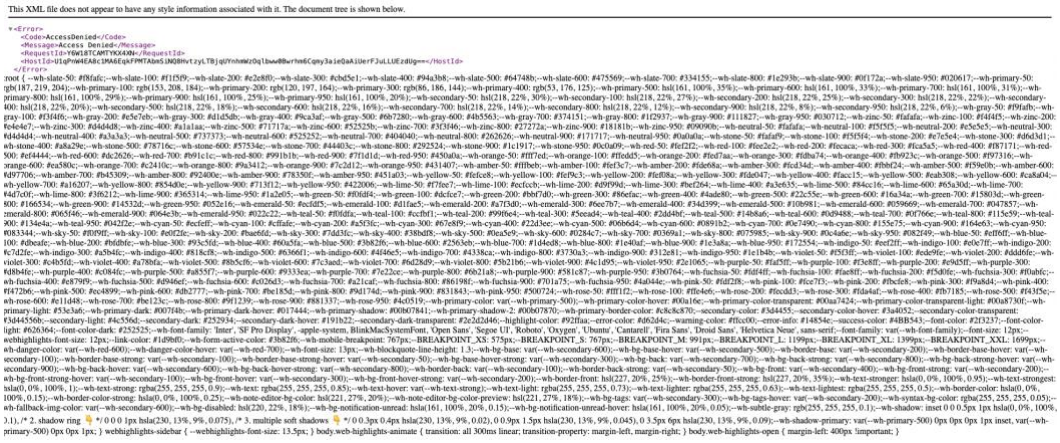

Figure 25: Denied direct access to S3 photo web page
URL: https://as2-ptantechasa-bucket.s3.amazonaws.com/cat2.jpeg

C. Auto Scaling Test

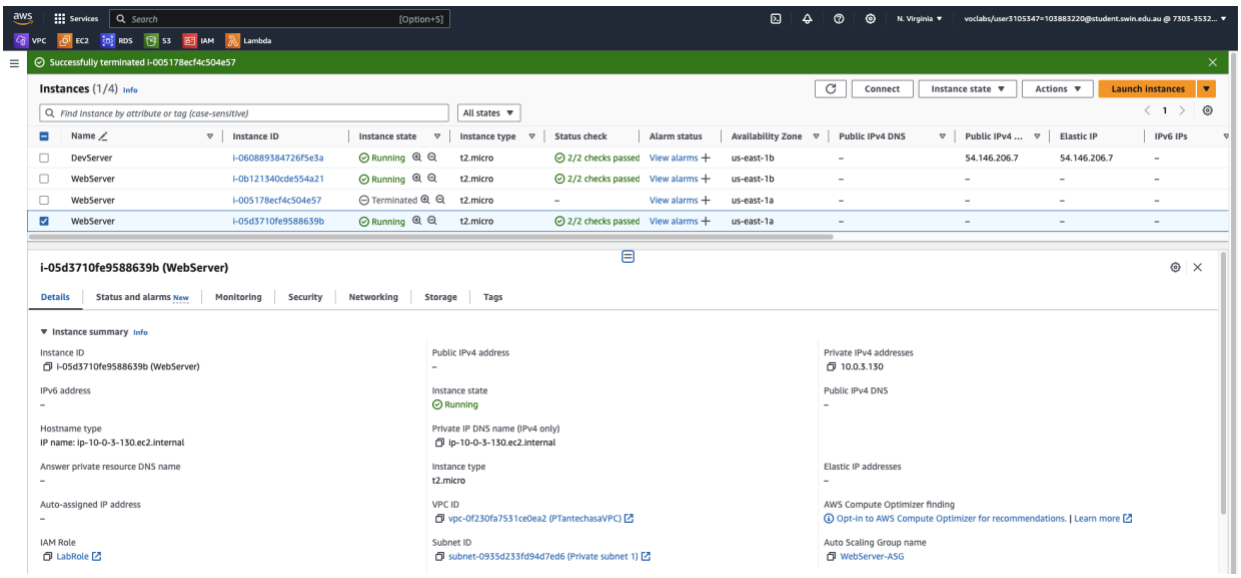

Figure 26: New Web Server instance had been created and fully functional when one instance is terminated

D. EC2 Targets Health

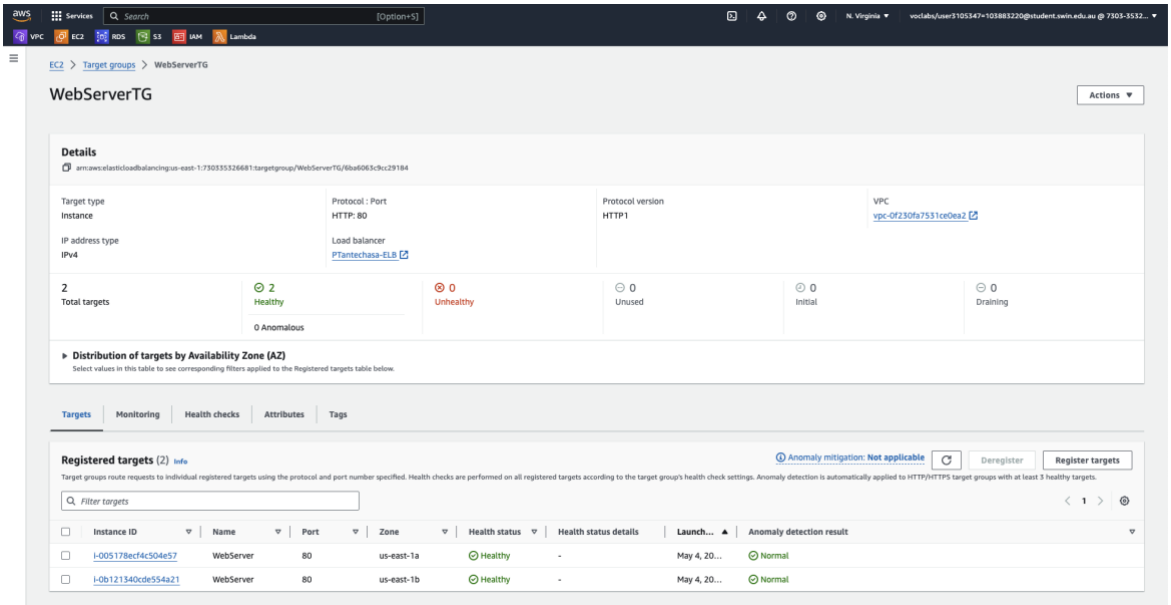

Figure 27: Evidence that both Web Server instance is Helathy

E. ICMP Traffic

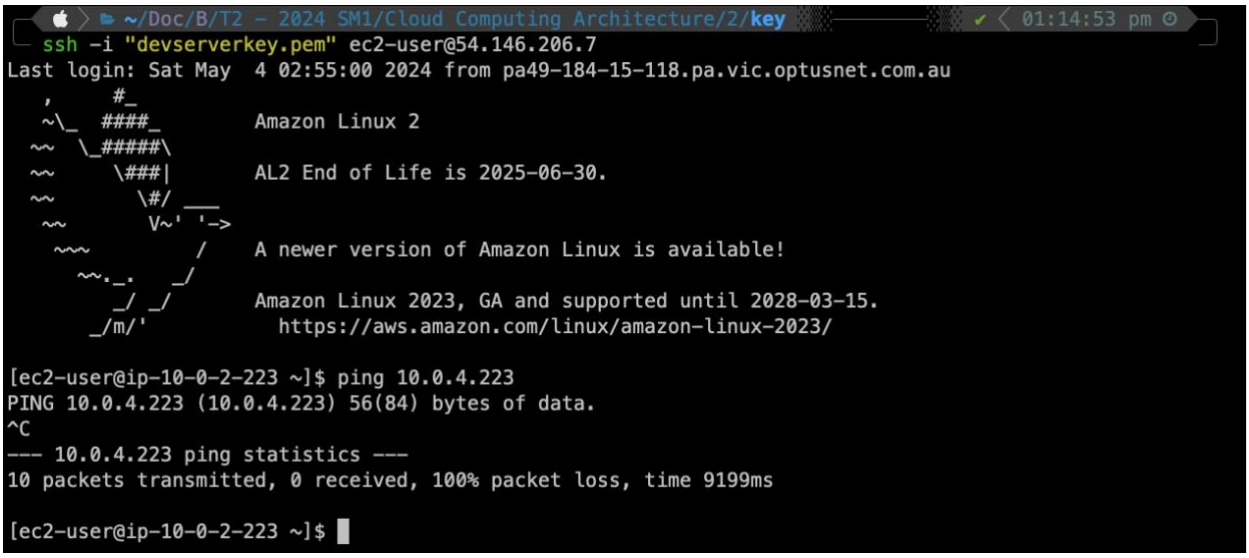

Figure 28: Unsuccessful ICMP communication between Dev and Web Server due to applied NACL.

REFERENCE

[1] AWS, "NAT Gateways," *Amazon.com*, 2020. https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-gateway.html